A Controlled Study of Fine-Tuning DINOv2 for Landmark Image Retrieval

Nguyen Ngoc Thanh Nhan, Tran Long Vu, Nguyen Hoang Anh Kiet, Nguyen Hai Nguyen, Nguyen Hong Quan, and Bui Cao Vu

FPT University, Da Nang, Vietnam
thanhnhan73.work@gmail.com, vutran.03092006@gmail.com, kiet.nguyen20067@gmail.com, hainguyenzzz2201@gmail.com, hongquannguyen2k6@gmail.com, vubc@fe.edu.vn

Abstract. Content-based image retrieval (CBIR) at the instance level asks for every database image that shows the *same* object as the query, under strong changes of viewpoint, scale, illumination and occlusion. Generalized-mean (GeM) pooling over fine-tuned CNN activations, trained without human annotation on structure-from-motion data, has been a standard global-descriptor baseline for this task since 2018. In this paper we revisit the GeM pipeline with a modern self-supervised Vision Transformer backbone (DINOv2 ViT-B/14) and improve it through a strictly controlled, one-factor-at-a-time protocol in which every new version changes exactly one design choice with respect to the best previous version and states its motivation in advance. Starting from a faithful rebuild of the GeM training recipe (contrastive loss, hard-negative mining on retrieval-SfM-120k, learned whitening), we study the descriptor head (GeM over patch tokens vs. the class token), the depth of partial fine-tuning, the learning rate, and α-weighted query expansion, and we select versions on the retrieval-SfM-120k validation split rather than on the test benchmarks. On the Revisited Oxford and Paris benchmarks, training on the class token instead of GeM-pooled patch tokens is the single largest gain (+7.8/+8.7 mAP on $\mathcal{R}$Oxford5k Medium/Hard over the GeM descriptor of the same backbone), while fine-tuning more blocks or lowering the learning rate gives equivalent models at about 72 mAP. Evaluated under exactly the same protocol, the fine-tuned class token improves over the frozen DINOv2 backbone by +1.9/+2.9 mAP on $\mathcal{R}$Oxford5k (72.0/52.8 vs. 70.1/49.8) and by +1.2/+1.1 on $\mathcal{R}$Paris6k; with αQE on both sides the final global-only model reaches 74.1/54.6 on $\mathcal{R}$Oxford5k and 90.2/77.7 on $\mathcal{R}$Paris6k, against 72.2/51.9 and 89.1/76.6 for the frozen backbone. Additional evaluation-only checks show that removing the ReLU from GeM does not close the gap to the class token, and that multi-scale inference hurts the class token only when it adds down-scaled inputs. The whole study runs on a single free-tier T4 GPU; code and logs are released.

Keywords: Content-based image retrieval · Instance-level retrieval · Landmark retrieval · GeM pooling · Vision Transformer · DINOv2 · Query expansion

1 Introduction

Content-based image retrieval (CBIR) ranks the images of a collection by their visual similarity to a query image, without relying on text or tags [35,36]. Its most demanding variant is *instance-level* retrieval, where a result is relevant only if it depicts the very same object or place as the query, for example a particular building photographed from a different side, at night, or partly hidden by trees [41,7]. Landmark retrieval is the canonical benchmark for this setting, and the Revisited Oxford and Paris datasets ($\mathcal{R}$Oxford5k, $\mathcal{R}$Paris6k) [27], with their Medium and Hard protocols, are the standard way to measure progress.

Over two decades the field moved from bag-of-visual-words on local SIFT features [34,21,25] to compact global descriptors produced by convolutional networks [33,4,3,38,13]. Among the latter, the method of Radenović *et al.* [29] became a reference point: a CNN is fine-tuned with a contrastive loss on image pairs mined automatically from 3D structure-from-motion (SfM) models, its last feature map is aggregated by *generalized-mean* (GeM) pooling with a learnable exponent, and a discriminative whitening is learned from the same SfM pairs. The recipe needs no manual labels, produces a single vector per image, and is still used as the global stage of state-of-the-art systems [5,20,32].

Since then, the backbone landscape has changed. Vision Transformers (ViTs) [11] trained with large-scale self-supervision, most notably DINO [6] and DINOv2 [23], produce features that already rival supervised retrieval models without any fine-tuning. This raises a natural, practical question that this paper answers experimentally: *what happens when the well-understood GeM training recipe is applied to a modern self-supervised ViT, and which parts of the recipe still matter?*

Rather than proposing a new architecture, we follow a disciplined engineering protocol. We first rebuild the GeM pipeline with only the backbone swapped (ResNet-101 $\rightarrow$ DINOv2 ViT-B/14), which we call B0. Each subsequent version B1, B2, ... changes *exactly one* factor with respect to the best version so far, and the reason for the change (prior result, theory and literature) is written down before the experiment is run. A version that does not improve on the validation split is not used as a base, and the next one branches from the previous best.

Our contributions are:

- A faithful, self-contained re-implementation of the GeM training and evaluation pipeline (SfM tuples, hard-negative mining, contrastive loss, learned whitening, revisited evaluation protocol) on top of DINOv2, runnable on a single free-tier GPU.
- A controlled one-factor-at-a-time study (B0–B4) of descriptor head, fine-tuning depth, learning rate and query expansion, with model and αQE selection on a held-out validation split, a frozen-backbone baseline run under the same protocol, and every intermediate number reported, including the negative results.
- The empirical finding that for a self-supervised ViT the class token is a much better trainable global descriptor than GeM over patch tokens (+7.8 mAP

on $\mathcal{R}$Oxford5k Medium) and that this gap is not caused by the ReLU in GeM; that multi-scale averaging with down-scaled inputs, which helps CNN descriptors, *hurts* the class token, while up-scaling does not; and that further fine-tuning capacity saturates at about 72 mAP.
– A global-only model reaching 74.1/54.6 mAP on $\mathcal{R}$Oxford5k and 90.2/77.7 on $\mathcal{R}$Paris6k (Medium/Hard), +1.9/+2.8 and +1.1/+1.1 above the frozen DINOv2 ViT-B/14 backbone with the same αQE and evaluation code.

The rest of the paper is organized as follows. Section 2 reviews related work. Section 3 describes the baseline, our rebuild and the versioning protocol. Section 4 gives the experimental setup, Section 5 the results, and Section 6 discusses findings and limitations before Section 7 concludes.

2 Related Work

From local features to global CNN descriptors. Classical instance retrieval quantized local SIFT descriptors [21] into visual words and ranked images with inverted files [34], refined by spatial verification [25,26] and aggregated encodings such as VLAD [17]. Razavian *et al.* [33] and Babenko *et al.* [4] showed that activations of ImageNet-trained CNNs are strong off-the-shelf descriptors. Pooling the last convolutional map then became the dominant design: sum pooling (SPoC) [3], max pooling (MAC) and regional max pooling (R-MAC) [38], and trainable aggregation layers such as NetVLAD [1]. Surveys [41,7,36] document this transition in detail.

Fine-tuning for retrieval. Off-the-shelf features are trained for classification, not for matching. Gordo *et al.* [13] fine-tuned R-MAC with a ranking loss on a cleaned landmark dataset, while Radenović *et al.* [28,29] removed the need for manual cleaning by mining positives and hard negatives from SfM reconstructions [31], introduced GeM pooling, and learned a discriminative whitening. Their contrastive loss [14] remains a strong choice, although later work trained on the much larger Google Landmarks v2 [39] with classification-style margin losses such as ArcFace [10].

Local features and re-ranking. DELF [22] learned attentive local features, DELG [5] unified local and global heads in one network, HOW [37] aggregated learned local descriptors, and DOLG [40] fused them orthogonally. Two-stage systems re-rank a global shortlist with local correlation (CVNet [20]); SuperGlobal [32] showed that improved global pooling and a cheap global re-ranking can match or exceed such systems. Query expansion [8,2], its weighted variant αQE [29] and diffusion [16] are training-free post-processing steps that exploit the neighbourhood of the query. Our study is deliberately *global-only*; αQE is the only post-processing we evaluate.

Transformers and self-supervised backbones. ViTs [11] were adapted to retrieval by El-Nouby *et al.* [12], who used the class token as the descriptor. Self-supervised ViTs, DINO [6] and DINOv2 [23], and language-supervised models such as CLIP [30] provide general-purpose features; DINOv2 reports strong $\mathcal{R}$Oxford5k/$\mathcal{R}$Paris6k results with a frozen backbone. Darcet *et al.* [9] observed high-norm artefact tokens in the patch outputs of large ViTs, and Kumar *et al.* [19] showed that full fine-tuning can distort pretrained features and hurt out-of-distribution transfer. Both observations turn out to be relevant to our findings.

3 Method

3.1 Problem Statement

Given a database $\mathcal{D} = \{I_1, \dots, I_N\}$ and a query image q (cropped to its bounding box, following the benchmark protocol), we learn an embedding $f : I \mapsto \mathbf{x} \in \mathbb{R}^D$, $\|\mathbf{x}\|_2 = 1$, and rank the database by cosine similarity $s(q, I_i) = \mathbf{x}_q^\top \mathbf{x}_i$. Only one global vector is stored per image, so search is a single matrix–vector product (or an approximate nearest-neighbour lookup at scale).

3.2 The GeM Baseline

GeM pooling. For a non-negative feature map with K channels and spatial locations $\mathcal{X}_k$ in channel k, GeM [29] computes

$$f_k = \Big(\frac{1}{|\mathcal{X}_k|} \sum_{x \in \mathcal{X}_k} x^p \Big)^{1/p}, \qquad k = 1, \dots, K, \tag{1}$$

which equals average pooling (SPoC) for $p = 1$ and tends to max pooling (MAC) as $p \to \infty$ (Fig. 1). The exponent p is learned jointly with the network and typically converges near 3.

Contrastive loss with SfM tuples. Training uses tuples $(q, p, n_1, \dots, n_M)$ with one positive and M hard negatives. For descriptors $\mathbf{x}_q$ and $\mathbf{x}_j$ and $d_j = \|\mathbf{x}_q - \mathbf{x}_j\|_2$, the loss is

$$\mathcal{L} = \tfrac{1}{2}\, d_p^2 \; + \; \sum_{m=1}^{M} \tfrac{1}{2} \max\big(0, \tau - d_{n_m}\big)^2, \tag{2}$$

with margin τ (0.85 for deep backbones [29]). Positives are image pairs that share enough 3D points in an SfM model; hard negatives are the nearest images in descriptor space that belong to a *different* 3D model, at most one per model, re-mined every epoch.

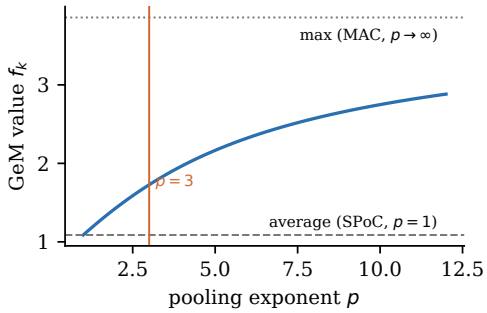


Fig. 1. Illustration of GeM pooling (Eq. 1) on a synthetic set of 64 non-negative activations: the pooled value moves continuously from the average ($p = 1$) towards the maximum ($p \to \infty$). Larger p focuses the descriptor on the strongest local responses.

Learned whitening (L_w). After training, a linear projection is learned from the descriptors of matching pairs $(i, j) \in \mathcal{P}$ mined from the SfM models [29]. Let $C_S = \frac{1}{|\mathcal{P}|} \sum_{(i,j)\in\mathcal{P}} (\mathbf{x}_i - \mathbf{x}_j)(\mathbf{x}_i - \mathbf{x}_j)^\top$ be the intra-class covariance, $\boldsymbol{\mu}$ the mean descriptor and C_D the covariance of non-matching pairs. The whitening is

$$P = \mathrm{eig}\big(C_S^{-1/2}\, C_D\, C_S^{-1/2}\big)^\top\, C_S^{-1/2}, \qquad \mathbf{x} \mapsto \ell_2\big(P(\mathbf{x} - \boldsymbol{\mu})\big), \tag{3}$$

where $\mathrm{eig}(\cdot)$ stacks eigenvectors in decreasing order of eigenvalue. As in the public implementation, C_D is estimated from the covariance of all training descriptors.

α-weighted query expansion. αQE [29] replaces the query by a weighted sum of itself and its top-n neighbours,

$$\mathbf{x}'_q = \ell_2\Big(\mathbf{x}_q + \sum_{i=1}^{n} \big(\mathbf{x}_q^\top \mathbf{x}_{(i)}\big)^\alpha\, \mathbf{x}_{(i)}\Big), \tag{4}$$

and searches again. Average query expansion [8] is the special case $\alpha = 0$.

3.3 B0: Rebuilding GeM on a Self-Supervised ViT

B0 keeps the complete GeM recipe and swaps only the backbone, from ResNet-101 [15] to DINOv2 ViT-B/14 [23]. An input image, resized so that its longer side is at most 518 pixels and cropped to a multiple of the patch size 14, yields one class token $\mathbf{t}_{\mathrm{cls}} \in \mathbb{R}^{768}$ and an $h \times w$ grid of patch tokens after the final LayerNorm. We reshape the patch tokens into a $768 \times h \times w$ map and apply

$$\mathbf{x}_{\mathrm{GeM}} = \ell_2\Big(\mathrm{GeM}_p\big(\mathrm{ReLU}(\mathbf{T}_{\mathrm{patch}})\big)\Big), \tag{5}$$

where the ReLU is required because LayerNorm outputs are signed while Eq. (1) assumes non-negative inputs. To keep the strong pretrained representation and

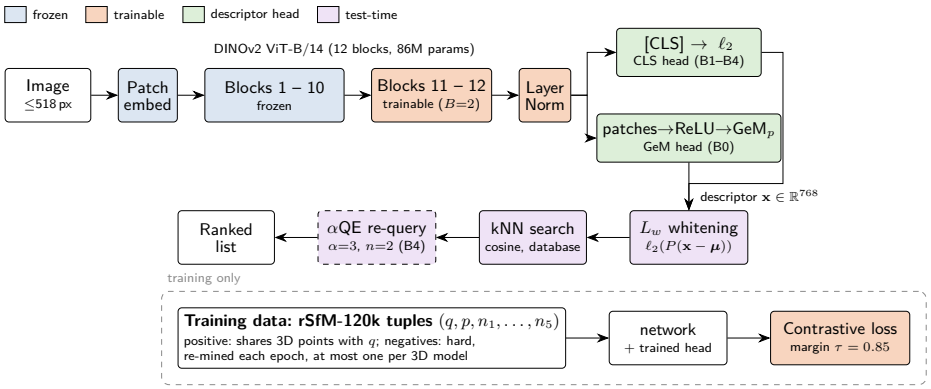


Fig. 2. Overview of the retrieval pipeline. DINOv2 ViT-B/14 tokens feed two descriptor heads: GeM over ReLU-rectified patch tokens (the original GeM design, trained in B0) and the ℓ_2-normalised class token (trained from B1 on). The descriptor is whitened with L_w, searched with cosine kNN and optionally expanded with αQE (B4). The dashed box shows training: the contrastive loss on SfM tuples updates only the last B blocks, the final LayerNorm and p (orange). In B0 the loss is attached to the GeM head, in B1–B4 to the CLS head.

fit a free GPU, only the last B of the 12 transformer blocks (plus the final LayerNorm and p) are trained; the first $12-B$ blocks run without gradient. With $B = 2$, about 14.2M of the 86M backbone parameters are trainable. Figure 2 summarises the pipeline.

Alongside the trained head, the network always exports two diagnostic descriptors from the same forward pass: the class token $\mathbf{x}_{\text{CLS}} = \ell_2(\mathbf{t}_{\text{cls}})$ and plain average pooling of patch tokens without ReLU, $\mathbf{x}_{\text{Avg}}$. They are evaluated with exactly the same whitening and protocol, which lets the data tell us which head to train next.

3.4 One-Factor-at-a-Time Improvement Protocol

Small benchmarks such as $\mathcal{R}$Oxford5k (70 queries) make it easy to over-interpret differences when several factors change at once. We therefore adopt three rules. (i) Each version changes *one* factor relative to the best version so far and keeps everything else, including the random seed. (ii) Before running, each version records *why* the change is expected to help: the previous result, the underlying theory, and a supporting paper. (iii) Versions are compared on a held-out *validation* task built from the rSfM-120k validation split (Section 4); $\mathcal{R}$Oxford5k and $\mathcal{R}$Paris6k are used only to report results. Differences smaller than about 1 validation mAP, which is within what a single seed can resolve, are treated as ties, and ties are kept in the lineage rather than resolved on the test sets. Test-time-only factors (such as αQE) do not require retraining; they are computed from the stored descriptors and reported as their own version, with and without the factor. Fig. 3 shows the resulting lineage.

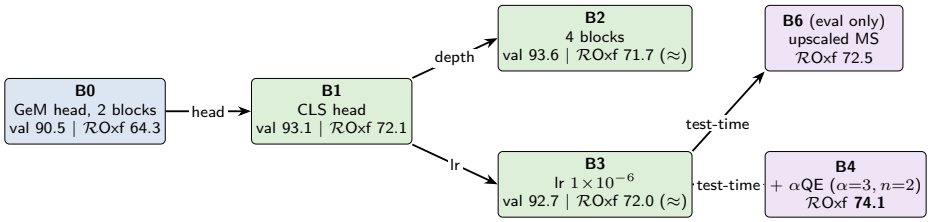


Fig. 3. Experimental lineage. Each arrow changes exactly one factor. "val" is the rSfM-120k validation mAP used for selection, $\mathcal{R}$Oxf the Medium mAP reported on the test set (trained head, single scale, L_w; B6: multi-scale with L_w). B1, B2 and B3 are within the tie margin (≈). B6 is an evaluation-only check on the B3 checkpoint; the planned ViT-L/14 version (B5) was not run.

In the first version of this study the branching decisions were taken on $\mathcal{R}$Oxf Medium mAP. We re-ran the selection on validation afterwards with the saved checkpoints (Table 3). It confirms the two decisions that matter, B0→B1 (train the class token) and adding αQE, but ranks B2 slightly above B1 and B3 (93.56 vs. 93.13 and 92.73 validation mAP). Since B1, B2 and B3 lie within 0.8 validation mAP and 0.5 test mAP of each other, we report them as equivalent and keep B3 as the base of B4, as in the original lineage; B2 + αQE is reported alongside it.

The versions and their motivation are:

B1 – train the CLS head instead of GeM. In B0 the untrained class token of the fine-tuned network outperformed the trained GeM head by +5.0/+4.0 mAP ($\mathcal{R}$Oxf M/H, single scale, L_w), and DINOv2 itself uses the class token for retrieval [23,12].

B2 – fine-tune 4 blocks instead of 2. Two of twelve blocks may lack the capacity to adapt to landmarks [12].

B3 – learning rate $2 \times 10^{-6} \to 1 \times 10^{-6}$ (from B1). B2 lowered the training loss but not test accuracy, suggesting the features drift away from the pretrained solution; a smaller step keeps the model closer to it [19], and [29] also used very small learning rates.

B4 – add αQE at test time (on B3). A cheap, training-free post-processing that is known to help on Oxford/Paris [29].

B6 – up-scaled multi-scale inference (on B3, evaluation only). B0 and B1 showed that the default scales $\{1, 1/\sqrt{2}, 1/2\}$ lower the class token (Section 5.2); if the cause is the loss of resolution at small scales, replacing them with $\{\sqrt{2}, 1, 1/\sqrt{2}\}$ should remove the drop.

4 Experimental Setup

Training data. We train on retrieval-SfM-120k [28,29], built from 3D models reconstructed from Flickr photos [31]. The training split contains 551 models

(about 133k images) and the validation split 162 models; models overlapping with the Oxford and Paris test landmarks were removed by the dataset authors. No manual labels are used.

Validation task. For model selection we build a retrieval task from the validation split: 1,000 query images and a database of 5,000 images drawn from the 162 validation models (fixed seed), where a database image is relevant if it belongs to the same 3D model as the query (on average 41.2 relevant images per query). We report validation mAP and mP@10 of each model's trained head with L_w, and tune the αQE parameters on a grid $\alpha \in \{0, 1, 3, 5, 10\}$, $n \in \{1, 2, 3, 5, 10\}$ on the same task.

Test data and metrics. We evaluate on $\mathcal{R}$Oxford5k and $\mathcal{R}$Paris6k [27] (4,993 and 6,322 database images, 70 queries each), the revisited versions of Oxford5k [25] and Paris6k [26]. Queries are cropped to their bounding box. We report mean average precision (mAP) under the Easy (E), Medium (M) and Hard (H) protocols, and mean precision at 10 (mP@10) where available. The 1M distractor set is not used.

Implementation details. All models use the DINOv2 ViT-B/14 weights from the official repository [23] and are implemented in PyTorch [24]. Table 1 lists the hyper-parameters, which follow the original GeM recipe wherever possible. Each epoch re-mines hard negatives for 2,000 random queries from a pool of 22,000 images, with 5 negatives per tuple; gradients are accumulated over 5 tuples per optimizer step. Training uses Adam [18] with exponential learning-rate decay ($\gamma = e^{-0.01}$ per epoch), weight decay 10^{-6}, a $10\times$ larger learning rate for p, and fp16 mixed precision. L_w is learned on 5,000 matching pairs from the rSfM-120k whitening split. Multi-scale (MS) descriptors average the descriptors at scales $\{1, 1/\sqrt{2}, 1/2\}$ (GeM-style power mean for the GeM head, arithmetic mean for the signed CLS/Avg descriptors); single scale (SS) uses scale 1 only. The frozen DINOv2 baseline is evaluated with exactly the same code, resolution, L_w (learned on its own descriptors) and αQE. Every run uses seed 0 and one NVIDIA T4 GPU (16 GB) on Kaggle; one epoch takes about 21 minutes with 2 trainable blocks and 23.5 minutes with 4, and a complete run (data download, 10 epochs, whitening and evaluation) takes 4.5–5 hours.

5 Results

5.1 Main Results

Table 2 reports every version with its trained descriptor, single scale and learned whitening, the frozen DINOv2 backbone evaluated under our protocol, and reference numbers taken from the literature. Figure 4 visualises the same numbers, and Table 3 gives the validation scores used for selection.

Four observations stand out. First, simply replacing ResNet-101 by DINOv2 ViT-B/14 in the GeM recipe (B0, GeM head) already improves $\mathcal{R}$Paris6k

Table 1. Training and evaluation settings. Only the highlighted factor changes between consecutive versions.

	Frozen	B0	B1	B2	B3	B4
Trained head	–	GeM	**CLS**	CLS	CLS	CLS
Fine-tuned blocks	0	2	2	**4**	2	2
Learning rate	–	2×10^{-6}	2×10^{-6}	2×10^{-6}	$\mathbf{1\times10^{-6}}$	1×10^{-6}
Query expansion	–	–	–	–	–	**αQE**

Shared by all versions: DINOv2 ViT-B/14; rSfM-120k; contrastive loss, $\tau = 0.85$; 10 epochs; 518 px (train/test); GeM $p_0 = 3$; Adam, weight decay 10^{-6}; L_w from 5k pairs; αQE with $\alpha = 3$, $n = 2$; seed 0.

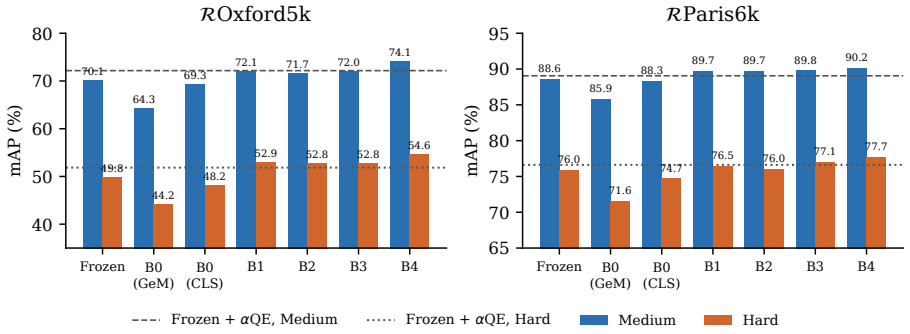


Fig. 4. mAP across versions (single scale, L_w), including the frozen DINOv2 backbone under our protocol. Dashed and dotted lines are the frozen backbone *with* αQE, the fair counterpart of B4. All fine-tuned class-token models (B1–B3) are on par with it on $\mathcal{R}$Oxford5k Medium and above it on Hard without any query expansion.

strongly (85.9/71.6 vs. 76.7/55.2) and $\mathcal{R}$Oxford5k Hard (44.2 vs. 40.1), but $\mathcal{R}$Oxford5k Medium stays at the ResNet level. Second, switching the trained descriptor from GeM to the class token (B1) is the largest single improvement: +7.8/+8.7 mAP on $\mathcal{R}$Oxford5k M/H over the B0 GeM head and +2.8/+4.7 over the untrained B0 class token, which shows that the gain comes both from the choice of descriptor and from training it directly. Third, B2 and B3 do not move the $\mathcal{R}$Oxford5k Medium score beyond the ±0.5 range around 72; only test-time αQE (B4) adds a further +2.1/+1.8 on $\mathcal{R}$Oxford5k and +0.3/+0.7 on $\mathcal{R}$Paris6k. Fourth, the comparison with the frozen backbone must be made under the same protocol. Evaluated with our code, the frozen DINOv2 class token with L_w reaches 70.1/49.8 on $\mathcal{R}$Oxford5k and 88.6/76.0 on $\mathcal{R}$Paris6k. Fine-tuning therefore helps, but by a moderate margin: +1.9/+2.9 (B3) on $\mathcal{R}$Oxford5k and +1.2/+1.1 on $\mathcal{R}$Paris6k without query expansion, and +1.9/+2.8 and +1.1/+1.1 when αQE is applied to both. Note that B0, which fine-tunes through the GeM head, is *below* the frozen class token; the benefit of fine-tuning appears only once the class token itself is trained. The published

Table 2. Main results (mAP, %) on the revisited benchmarks. Our rows use single-scale descriptors with L_w; Δ is the change in $\mathcal{R}$Oxf Medium w.r.t. the version each row branches from; "val" is the rSfM-120k validation mAP. Rows with and without αQE are grouped so that each is compared with its counterpart. Reference rows are copied from the cited sources and differ in protocol (†: multi-scale with whitening, from the official cnnimageretrieval-pytorch release of [29]; ‡: frozen backbone, Table 9 of [23]).

			$\mathcal{R}$Oxford5k			$\mathcal{R}$Paris6k			
Version	Descriptor	val	E	M	H	E	M	H	ΔM
Reference results (other protocols)									
GeM R101† [29]	GeM	–	–	65.4	40.1	–	76.7	55.2	
DINOv2 ViT-B/14‡ [23]	CLS	–	–	72.9	49.5	–	90.3	78.6	
Ours, no query expansion (DINOv2 ViT-B/14, global descriptor only)									
Frozen DINOv2	CLS	90.96	85.15	70.06	49.84	94.24	88.64	75.96	–
B0	GeM (trained)	90.50	77.71	64.30	44.17	93.05	85.87	71.60	–
B0	CLS (not trained)	91.13	83.76	69.28	48.19	93.76	88.30	74.73	
B1: CLS head	CLS	93.13	**85.71**	**72.10**	**52.88**	95.53	89.72	76.46	+7.80
B2: 4 blocks	CLS	**93.56**	85.01	71.67	52.77	95.21	89.70	76.04	−0.43
B3: lr 10^{-6}	CLS	92.73	85.64	71.99	52.77	95.17	**89.83**	**77.06**	−0.11
Ours, with αQE ($\alpha = 3$, $n = 2$)									
Frozen DINOv2 + αQE	CLS	–	87.33	72.17	51.85	94.45	89.06	76.62	–
B2 + αQE	CLS	–	87.06	73.61	**54.68**	95.30	89.88	76.31	+1.94
B4: B3 + αQE	CLS	–	**87.91**	**74.09**	54.60	**95.58**	**90.17**	**77.73**	+2.10

Table 3. Selection on the rSfM-120k validation task (1,000 queries, 5,000 database images; trained head, single scale). Loss: contrastive loss of the trained head on validation tuples. Bold: best.

	Frozen	B0 (GeM)	B0 (CLS)	B1	B2	B3
val mAP	87.81	88.11	88.62	91.41	**91.62**	90.61
val mAP + L_w	90.96	90.50	91.13	93.13	**93.56**	92.73
val mP@10 + L_w	98.10	98.20	98.55	98.89	**99.06**	98.83
val loss	0.142	0.109	0.113	0.092	**0.081**	0.098

DINOv2 figures [23] (72.9/49.5 and 90.3/78.6) were obtained with a different evaluation protocol and are listed for reference only.

The validation task (Table 3) agrees with the test sets on the large effects: training the class token (B1) gains +2.6 validation mAP over the B0 GeM head and +2.2 over the frozen backbone. It ranks B2 first, but the spread among B1–B3 (0.8 mAP) is of the same size as their spread on the test sets (0.4 mAP on $\mathcal{R}$Oxf Medium), and the two orderings differ, so we do not read a winner into it. Validation-selected B2 with αQE gives 73.6/54.7 on $\mathcal{R}$Oxford5k and 89.9/76.3 on $\mathcal{R}$Paris6k, within 0.5 mAP of B4.

5.2 Descriptor Head, Whitening and Scale

Table 4 and Fig. 5 break down all descriptor variants exported by B0 and B1. Learned whitening consistently helps on $\mathcal{R}$Oxford5k (e.g. +4.7 mAP Medium

Table 4. Descriptor heads of B0 and B1 on $\mathcal{R}$Oxford5k / $\mathcal{R}$Paris6k (mAP Medium/Hard). SS/MS: single/multi-scale; L_w: learned whitening. Bold: best per model.

		B0 (GeM head trained)				B1 (CLS head trained)			
Head	Setting	$\mathcal{R}$Oxf M	H	$\mathcal{R}$Par M	H	$\mathcal{R}$Oxf M	H	$\mathcal{R}$Par M	H
GeM	SS	61.27	43.03	86.09	71.99	37.26	12.07	71.63	48.97
	SS + L_w	64.30	44.17	85.87	71.60	55.85	30.47	83.34	67.27
	MS	62.58	44.35	86.43	72.66	39.57	14.29	74.97	53.84
	MS + L_w	65.15	45.31	87.21	73.33	55.71	29.74	83.70	67.87
CLS	SS	64.59	45.21	87.33	73.49	70.41	50.51	88.99	75.74
	SS + L_w	**69.28**	48.19	88.30	74.73	**72.10**	**52.88**	**89.72**	**76.46**
	MS	65.41	48.22	87.69	74.22	67.79	47.19	88.74	75.45
	MS + L_w	68.45	**48.98**	**88.63**	**75.45**	69.82	50.77	89.53	76.36
Avg	SS	62.08	44.55	85.62	71.47	36.18	11.99	71.98	50.21
	SS + L_w	65.81	45.60	86.86	73.16	54.14	26.71	82.91	66.85
	MS	62.54	45.04	86.07	72.13	38.33	14.22	75.23	55.14
	MS + L_w	64.86	45.51	87.26	73.68	53.81	26.42	83.10	67.41

for the B0 class token), while its effect on $\mathcal{R}$Paris6k is small. In B0, multi-scale averaging gives the usual small gain to the GeM head (+0.9 M) but already reduces the class token (69.3→68.5). Once the class token is trained (B1), multi-scale hurts it by 2.3 mAP Medium; we therefore use single scale for all CLS results. The B6 check (Table 5, bottom) shows that the drop comes from the down-scaled inputs and not from averaging itself: on the B3 checkpoint, replacing $\{1, 1/\sqrt{2}, 1/2\}$ by $\{\sqrt{2}, 1, 1/\sqrt{2}\}$ turns a −2.1 mAP loss into a +0.5 gain on $\mathcal{R}$Oxf Medium (72.5 vs. 72.0; 75.1 vs. 74.1 with αQE), while $\mathcal{R}$Paris6k is unchanged. The gain is small and comes at about twice the extraction cost, so single scale remains our default. Finally, in B1 the GeM and Avg heads, which are no longer trained, collapse (to about 55 M / 30 H on $\mathcal{R}$Oxford5k with L_w), confirming that the last blocks have specialised to the class-token objective.

5.3 Fine-Tuning Depth and Learning Rate

Figure 6 plots the training loss of B1–B3. Fine-tuning four blocks (B2) lowers the final contrastive loss by 24% (0.076 vs. 0.099) and also gives the best validation loss and mAP (Table 3), yet yields slightly *lower* test accuracy on both datasets (Table 2). Halving the learning rate (B3) leaves a higher final loss (0.116), a lower validation mAP and the same $\mathcal{R}$Oxford5k accuracy (72.0/52.8), with a small gain on $\mathcal{R}$Paris6k Hard (+0.6). Training and validation, which come from the same SfM distribution, move together, but the test benchmarks do not follow: after B1, better fitting rSfM-120k no longer transfers to Oxford and Paris.

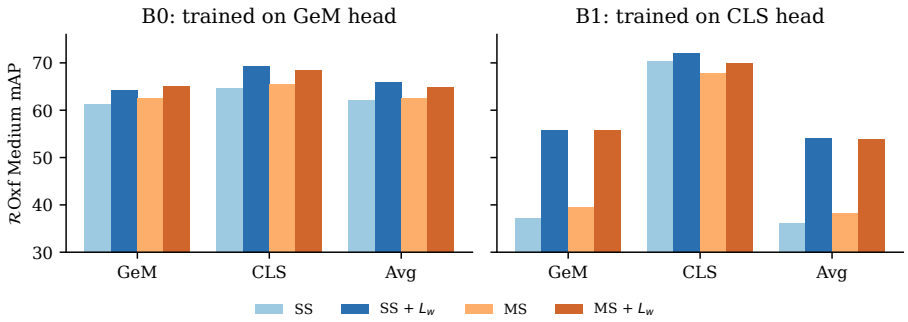


Fig. 5. $\mathcal{R}$Oxf Medium mAP of the three descriptor heads under four test settings. Left: B0, trained through the GeM head; right: B1, trained through the class token. Training the class token improves it and makes the other two heads collapse; multi-scale averaging lowers the class token in both models.

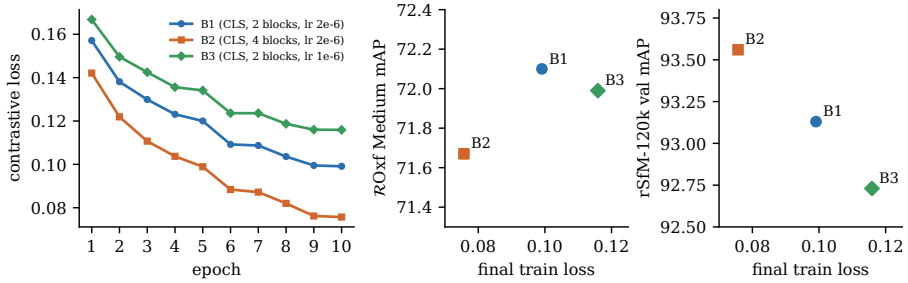


Fig. 6. Left: training loss per epoch for B1–B3 (from the training logs). Middle: final training loss versus $\mathcal{R}$Oxf Medium mAP. Right: final training loss versus rSfM-120k validation mAP. B2 fits the training data best and is best on validation, but retrieves slightly worse on the test set.

5.4 Query Expansion

Table 6 isolates the effect of αQE on the B3 model. With $\alpha = 3$ and $n = 2$, αQE improves every metric on $\mathcal{R}$Oxford5k, including precision at 10 (+1.9 M, +3.6 H), and gives smaller but consistent gains on $\mathcal{R}$Paris6k mAP. The improvement holds for both single- and multi-scale descriptors, but single scale remains the better choice.

The setting $\alpha = 3$, $n = 2$ is the one commonly used in the literature [29]. Tuning it on the validation task instead selects $\alpha = 0$, $n = 10$, i.e. plain average query expansion over ten neighbours (94.71 vs. 93.96 validation mAP for B2 with $\alpha = 3$, $n = 2$; the trend is the same for the frozen backbone and B3). This choice does not transfer uniformly (last row of Table 6): it raises $\mathcal{R}$Paris6k to 91.1/79.3 but lowers $\mathcal{R}$Oxford5k to 69.8/50.1, below no expansion at all. The validation task has about 41 relevant images per query, so expanding with many

Table 5. Evaluation-only checks (mAP Medium/Hard, L_w). Top: pooling variants on the B0 checkpoint (trained through standard GeM), single scale. GeM-signed removes the ReLU and applies a signed power mean; GeM-shift subtracts the per-channel minimum before GeM. Bottom: multi-scale variants on the B3 checkpoint (class token).

		$\mathcal{R}$Oxf (no QE)		$\mathcal{R}$Par (no QE)		$\mathcal{R}$Oxf + αQE	
Model	Variant	M	H	M	H	M	H
B0	GeM (ReLU, trained)	64.31	44.20	85.87	71.60	–	–
B0	GeM-signed (no ReLU)	64.86	45.00	86.90	73.10	–	–
B0	GeM-shift ($x - \min$)	52.90	33.81	79.49	62.49	–	–
B0	Avg (no ReLU)	65.81	45.60	86.86	73.17	–	–
B0	CLS (not trained)	**69.29**	**48.19**	**88.30**	**74.74**	–	–
B3	SS: $\{1\}$	71.99	52.77	89.83	77.06	74.09	54.60
B3	MS: $\{1, 1/\sqrt{2}, 1/2\}$	69.88	50.44	89.57	76.72	72.49	53.58
B3	B6, MS-up: $\{\sqrt{2}, 1, 1/\sqrt{2}\}$	**72.49**	**53.57**	89.80	**77.12**	**75.09**	**55.44**

Table 6. Effect of αQE on the B3 model, CLS descriptor with L_w. B4 = B3 + αQE ($\alpha = 3$, $n = 2$), single scale. The last row uses the setting chosen on validation ($\alpha = 0$, $n = 10$).

		$\mathcal{R}$Oxford5k				$\mathcal{R}$Paris6k			
Scale	QE	M	H	mP@10 M	mP@10 H	M	H	mP@10 M	mP@10 H
SS	–	71.99	52.77	86.43	66.11	89.83	77.06	98.00	89.00
SS	αQE	**74.09**	**54.60**	**88.29**	**69.68**	90.17	77.73	**98.14**	89.00
MS	–	69.88	50.44	84.81	61.41	89.57	76.72	98.00	88.43
MS	αQE	72.49	53.58	86.29	66.41	89.89	77.24	97.86	88.14
SS	αQE (0, 10)	69.76	50.11	84.14	67.14	**91.09**	**79.26**	98.00	**89.14**

neighbours is safe there; $\mathcal{R}$Oxford5k has far fewer relevant images per query and many hard negatives, so large n pulls in false positives. We therefore keep $\alpha = 3$, $n = 2$ as the main setting and report the validation choice for transparency; a validation set whose statistics are closer to the test landmarks would be needed to tune αQE reliably.

6 Discussion

Why does the class token beat GeM on a ViT? GeM was designed for CNN feature maps whose post-ReLU activations are non-negative and spatially sparse, so that a power mean highlights discriminative regions. ViT patch tokens after LayerNorm are dense and signed, so a ReLU has to be inserted before GeM. A natural hypothesis is that this ReLU throws away useful negative evidence. Our ablation (Table 5, top) does not support it as the main cause: a signed power mean without ReLU gains only +0.6/+0.8 mAP on $\mathcal{R}$Oxford5k (+1.0/+1.5 on $\mathcal{R}$Paris6k) over standard GeM, and plain average pooling, which also keeps

the sign, behaves the same; both remain 3.5–5 mAP below the untrained class token. Making the tokens non-negative by shifting instead of clipping is clearly harmful (-11.4 mAP), because the shift is dominated by a few extreme values. The gap is therefore a property of *what* is pooled rather than of the ReLU: any uniform pooling over patch tokens, including the few high-norm artefact tokens that large self-supervised ViTs place in background regions [9], mixes in non-discriminative content, whereas the class token is precisely the output that DINOv2's self-distillation objective shapes into a global image summary [6,23]. The learned GeM exponent in B0 stayed close to its initial value ($p = 2.956$), which also suggests that the GeM head did not find a better operating point on these tokens. We did not test the artefact-token explanation directly (e.g. with register tokens), so it remains a plausible but unverified account.

Why does multi-scale hurt the class token? For CNNs, multi-scale averaging adds robustness to object size. A ViT trained at a fixed resolution sees fewer patches at scales $1/\sqrt{2}$ and $1/2$ (as few as 18×18 tokens), and its position embeddings must be interpolated; the class token at small scales therefore summarises a coarser, less discriminative view, and averaging it with the full-scale token dilutes the descriptor. The B6 check supports this explanation: with the same averaging but scales $\{\sqrt{2}, 1, 1/\sqrt{2}\}$, the drop disappears and $\mathcal{R}$Oxford5k gains $+0.5/+0.8$ mAP over single scale (Table 5). The scale at $1/\sqrt{2}$ is still included in B6, so the result points at the smallest scale ($1/2$) as the main source of the loss.

How much does fine-tuning help? Under the same protocol the gain of fine-tuning over the frozen DINOv2 class token is real but moderate (about $+2/+3$ mAP on $\mathcal{R}$Oxford5k and $+1$ on $\mathcal{R}$Paris6k), and it exists only when the class token itself is trained: B0, trained through GeM, ends below the frozen backbone. The strong frozen baseline is consistent with DINOv2's design as a general-purpose feature extractor [23], and it means that, for this backbone, the choice of descriptor matters more than the amount of fine-tuning.

Saturation of fine-tuning. B1–B3 all land within 0.5 mAP of each other on $\mathcal{R}$Oxford5k Medium, and the B2 experiment shows that more trainable capacity improves the training and validation objectives without improving retrieval on the test landmarks. This agrees with the observation that fine-tuning can distort pretrained features and hurt transfer to a shifted distribution [19]: rSfM-120k is relatively small and visually different from the Oxford and Paris landmarks, so beyond a light adaptation the model starts to specialise to the training set. The remaining levers are therefore a stronger backbone (e.g. ViT-L/14), a larger and more diverse training set such as GLDv2 [39] with margin losses [10], or post-processing and re-ranking [16,32].

Limitations. The study is limited by a free GPU budget of roughly 30 hours per week, which shaped several choices. (i) Each configuration was trained once with seed 0; we did not measure run-to-run variance, so differences below about 1 mAP (B1 vs. B2 vs. B3, on validation or test) should be read as ties, and the

fine-tuning gain over the frozen backbone (about 2 mAP on $\mathcal{R}$Oxford5k) should be confirmed with more seeds. (ii) The branching decisions were originally taken on $\mathcal{R}$Oxford5k; the validation selection was carried out afterwards with the saved checkpoints and agrees on the large effects but not on the ranking of B1–B3. (iii) Our validation task has many more relevant images per query than $\mathcal{R}$Oxford5k, which made it unreliable for tuning αQE. (iv) The ViT-L/14 version (B5) was not run, and B6 and the GeM ablations are evaluation-only checks on existing checkpoints rather than retrained models. (v) We do not evaluate with the $\mathcal{R}$1M distractors, and we do not compare against two-stage systems with local-feature re-ranking (e.g. CVNet [20]) or against SuperGlobal [32], which operate in a different cost regime. The external reference numbers in Table 2 use different protocols and are indicative only; our own frozen-backbone rows are the fair baseline.

7 Conclusion

We revisited the GeM landmark-retrieval recipe with a self-supervised DINOv2 ViT-B/14 backbone and improved it through a transparent one-factor-at-a-time protocol, with selection on a validation split and a frozen baseline run under the same protocol. The study shows that, on a ViT, the class token rather than GeM pooling of patch tokens is the descriptor worth training, and that the gap is not caused by the ReLU in GeM; that multi-scale averaging hurts the class token when it adds down-scaled inputs, but not when it up-scales; and that once the right descriptor is trained, neither more trainable blocks nor a smaller learning rate helps. Fine-tuning the class token on rSfM-120k improves over the frozen DINOv2 backbone by about +2/+3 mAP on $\mathcal{R}$Oxford5k and +1 on $\mathcal{R}$Paris6k. With αQE added at test time, the resulting global-only model reaches 74.1/54.6 mAP on $\mathcal{R}$Oxford5k and 90.2/77.7 on $\mathcal{R}$Paris6k, against 72.2/51.9 and 89.1/76.6 for the frozen backbone with the same αQE. Future work includes repeated seeds, a larger backbone, training on GLDv2, a validation set closer to the test landmarks, and efficient global re-ranking.

Code and data availability. All notebooks, source code, raw result files and training and evaluation logs used in this paper (B0–B4, the frozen baseline, the validation selection and the evaluation-only checks) are included in the accompanying repository. The fine-tuned weights are available from the authors on request. The datasets are publicly available from their original authors [29,27].

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